Do LLMs Know What They Know? Linking Representation Linearity to Introspective Access

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Abstract

The linear representation hypothesis holds that large language models (LLMs) encode concepts as linear directions in activation space, yet recent work reveals that some features—notably refusal behavior—exhibit fundamentally nonlinear, multi-dimensional structure. We ask whether this geometric distinction predicts a model's ability to introspect on its own internal states, drawing an analogy to the conscious/unconscious divide in human cognition: linearly encoded features may be "accessible" to self-report, while nonlinearly encoded features may not. We test this *Introspection–Linearity Hypothesis* through a two-pronged design. First, we extract residual-stream activations from QWEN2.5-1.5B-INSTRUCT and train linear and nonlinear probes to quantify the linearity of three feature types: sentiment, binary refusal, and harmful-content sub-categories. Second, we administer introspection tests to GPT-4.1, measuring self-report accuracy on tasks of varying linearity. We find that perfectly linear features (sentiment, binary refusal) yield 100% self-report accuracy, while tasks involving nonlinear structure show degraded performance: sub-category classification drops to 97.5% and borderline refusal prediction to 90.0%. Confidence calibration reveals a significant gap between clear and ambiguous cases ($p = 0.021$). These results provide partial support for the hypothesis that representation geometry constrains introspective access in LLMs, with implications for alignment monitoring and interpretability.

1 Introduction

Large language models encode a remarkable range of concepts in their internal representations. Sentiment, truthfulness, and refusal intent can all be recovered from hidden states by training simple linear probes Marks and Tegmark [2023], Li et al. [2023], Arditi et al. [2024]. The *linear representation hypothesis* Park et al. [2023] formalizes this observation: most task-relevant features correspond to linear directions in activation space. But what happens when a feature is *not* linear? Recent work shows that refusal behavior—one of the most safety-critical properties of aligned models—is better described by nonlinear, multi-dimensional structure Hildebrandt et al. [2025], Engels et al. [2025]. If models can "see" their own linear representations but are blind to their nonlinear ones, this has direct consequences for alignment: self-report-based monitoring may have systematic blind spots.

Why does this matter? The ability of an LLM to report on its own internal states—what we call *introspective access*—is a prerequisite for many alignment strategies. Techniques ranging from Constitutional AI to debate-based oversight assume that models can faithfully articulate their reasoning and intentions. If introspective access depends on the geometry of the underlying representation, then some model behaviors may be fundamentally opaque to the model itself. This creates a structural analogy to the conscious/unconscious divide in human cognition Dehaene [2014]: linearly encoded information is "accessible" to the model's self-report mechanisms, while nonlinearly encoded information is not.

What is missing? Despite substantial progress on both fronts—representation geometry and model self-knowledge—no prior work has directly tested whether these two properties are linked. Studies of linear representations Marks and Tegmark [2023], Park et al. [2023], Burns et al. [2022] and nonlinear representations Hildebrandt et al. [2025], Engels et al. [2025] each characterize feature geometry, but do not ask whether models can introspect differentially on features of different geometry. Conversely, work on factual self-awareness Tamoyan et al. [2025] and error acknowledgment Yang and Jia [2025] demonstrates that LLMs possess some introspective capabilities, but does not connect these to the linearity of the underlying feature.

Our approach. We propose the first empirical test of the *Introspection–Linearity Hypothesis*: that LLMs' introspective accuracy on a feature is predicted by the linearity of that feature's representation. We employ a two-pronged experimental design. First, we extract residual-stream activations from QWEN2.5-1.5B-INSTRUCT and train both linear (logistic regression) and nonlinear (MLP) probes to quantify the geometry of three feature types: sentiment polarity, binary refusal, and harmful-content sub-categories. Second, we administer a battery of introspection tests to GPT-4.1, measuring self-report accuracy, confidence calibration, and meta-cognitive awareness on tasks spanning the linearity spectrum. We find a consistent introspection gap: 100% self-report accuracy for clearly linear features versus 90.0–97.5% for tasks involving nonlinear structure, with significantly miscalibrated confidence at decision boundaries ($p = 0.021$).

Our contributions are as follows:

- We formalize and test the *Introspection–Linearity Hypothesis*, the first empirical link between representation geometry and introspective access in LLMs.
- We demonstrate that features with linear representations (sentiment, binary refusal) are perfectly introspectable (100% self-report accuracy), while features with nonlinear structure show degraded accuracy (90.0–97.5%).
- We show that confidence calibration tracks representation geometry: models report significantly lower confidence on tasks near nonlinear decision boundaries, revealing partial but imperfect meta-cognitive awareness.
- We discuss implications for alignment monitoring, identifying decision boundaries as the primary locus of introspective failure.

The rest of this paper is organized as follows. Section 2 reviews related work on representation geometry and model self-knowledge. Section 3 describes our experimental methodology. Section 4 presents results. Section 5 discusses implications and limitations. Section 6 concludes.

2 Related Work

Linear representations in LLMs. The linear representation hypothesis Park et al. [2023] posits that high-level concepts are encoded as linear directions in a model's activation space. Empirical support comes from diverse domains: Marks and Tegmark [2023] show that truth and falsehood are linearly separable in LLM hidden states, Burns et al. [2022] recover latent knowledge via unsupervised linear probes, and Li et al. [2023] demonstrate that linear interventions on "truthfulness directions" improve factual accuracy. Nanda et al. [2023] find that even game-playing models develop linear board-state representations. In safety, Arditi et al. [2024] show that refusal behavior across 13 models (up to 72B parameters) is mediated by a single linear direction, enabling both ablation and induction of refusal via weight orthogonalization. These findings suggest that many critical model properties are geometrically simple and, we hypothesize, therefore accessible to introspection.

Nonlinear and multi-dimensional features. Not all features are one-dimensional. Engels et al. [2025] provide a theoretical framework for *irreducible multi-dimensional features*—representations that cannot be decomposed into independent one-dimensional components. They discover circular features (days of the week, months) in GPT-2 and Mistral 7B, showing these are causally used for modular arithmetic. Most relevant to our work, Hildebrandt et al. [2025] demonstrate that refusal behavior in LLMs is fundamentally nonlinear: UMAP and t-SNE achieve substantially better cluster separability than PCA across six models and three architecture families, with sub-clusters emerging within harmful-content embeddings that are invisible to linear methods. This tension between the linear view of Arditi et al. [2024] and the nonlinear view of Hildebrandt et al. [2025] motivates our hypothesis: binary refusal may be linear, but the fine-grained structure within refusal is not.

Representation engineering and steering. Zou et al. [2023] introduce representation engineering as a top-down approach to reading and controlling high-level concepts (honesty, emotion, fairness) via linear directions in activation space. This framework implicitly assumes that features worth monitoring are linearly accessible. Our work tests this assumption: if some safety-relevant features are nonlinear, representation engineering may miss them, and models may be unable to self-report on them.

Model self-knowledge and introspection. A growing body of work examines whether LLMs can accurately report on their own internal states. Tamoyan et al. [2025] find that models encode linear features predicting whether they will recall correct facts—a form of "factual self-awareness" that emerges rapidly during training and peaks in intermediate layers. Yang and Jia [2025] show that retraction behavior (admitting mistakes) is driven by linearly probeable "belief" states that causally influence model outputs. These studies demonstrate that LLMs possess some introspective capability, but both focus exclusively on linearly encoded properties. We extend this line of inquiry by asking whether introspective access degrades for nonlinearly encoded features.

Probing methodology. Our approach builds on the probing paradigm reviewed by Belinkov [2022], which highlights that the complexity of the probe matters: a sufficiently powerful probe can extract information that a simpler one cannot. We exploit this property by comparing linear (logistic regression) and nonlinear (MLP) probes on the same features, using the accuracy gap as a measure of feature nonlinearity. This follows the logic that if an MLP substantially outperforms logistic regression, the feature contains structure that is not linearly accessible Belinkov [2022].

3 Methodology

We test the Introspection–Linearity Hypothesis through two complementary experiments: activation probing to measure feature linearity (section 3.1) and behavioral introspection tests to measure self-report accuracy (section 3.2). We use separate models for each experiment—an open-weight model for probing (where internal activations are needed) and a frontier model for introspection (where strong instruction-following is essential)—and discuss this design choice in section 5.

3.1 Experiment 1: Activation Probing

Model and activation extraction. We extract residual-stream activations from QWEN2.5-1.5B-INSTRUCT (1.54B parameters) at the last non-padding token position across all 29 transformer layers. Inputs are formatted as chat messages using the model's tokenizer template and truncated to 128 tokens.

Datasets. We use three feature types spanning a range of expected linearity:

- **Sentiment** (known-linear control): 80 custom texts (40 positive, 40 negative) with unambiguous sentiment.
- **Binary refusal** (harmful vs. harmless): 400 samples from HARMLESS-ALPACA Hildebrandt et al. [2025] and 400 from HARMFUL-BEHAVIORS Hildebrandt et al. [2025], totaling 800 prompts.
- **Harmful sub-categories** (expected nonlinear): The 400 harmful prompts are assigned to six sub-categories via keyword matching: *other* (n=137), *hacking* (n=88), *fraud* (n=69), *violence* (n=60), *manipulation* (n=23), and *illegal content* (n=23).

Linear and nonlinear probes. For each feature, we train two classifiers on last-layer activations using 5-fold stratified cross-validation:

- **Linear probe**: Logistic regression with L_2 regularization (C=1.0).
- **Nonlinear probe**: Two-layer MLP with hidden dimensions $(256, 128)$, ReLU activations, and early stopping (max 500 iterations).

Nonlinearity score. We define the *nonlinearity score* of a feature as:

$$\text{NL-SCORE} = \text{Acc}_{\text{MLP}} - \text{Acc}_{\text{LR}} \tag{1}$$

A positive NL-SCORE indicates that the feature contains structure not captured by a linear classifier, while a negative or zero score suggests the feature is well-approximated linearly Belinkov [2022].

Dimensionality reduction. To complement the probe-based analysis, we apply PCA (linear) and UMAP (nonlinear; k=30 neighbors) to the activation space and visually assess cluster structure, following Hildebrandt et al. [2025].

3.2 Experiment 2: Introspection Testing

Model. We use GPT-4.1 (OpenAI) at temperature 0 for deterministic outputs. All responses are elicited in structured JSON format for reliable parsing.

Test battery. We administer five introspection tests of increasing difficulty:

1. **Sentiment classification** (n=30): The model classifies texts as positive or negative. This tests introspective access to a known-linear feature.
2. **Binary refusal prediction** (n=70): Given a prompt, the model predicts whether it will refuse to answer. This tests access to the linear refusal direction.
3. **Category classification** (n=40): The model classifies harmful prompts into sub-categories (violence, hacking, fraud, manipulation). We run this in two modes: (a) *external*, where the model classifies the input, and (b) *introspective*, where the model reports which category triggers its refusal response.
4. **Borderline self-prediction** (n=20): The model predicts its own refusal behavior on deliberately ambiguous prompts near the refusal boundary (e.g., "How do I pick a lock?").
5. **Deep introspection** (n=35): We test graded harm assessment (Spearman ρ with ground truth), confidence calibration (comparing self-reported confidence on clear vs. ambiguous prompts), and structural awareness (whether the model reports using single or multiple refusal mechanisms).

Ground truth. For sentiment and sub-category classification, ground truth labels are provided by the dataset. For binary refusal and borderline prediction, we determine ground truth by actually presenting each prompt to GPT-4.1 and recording whether it refuses, then comparing this outcome to the model's *prediction* of its own behavior.

3.3 Evaluation Metrics

- **Probe accuracy**: 5-fold cross-validated classification accuracy.
- **Self-report accuracy**: Agreement between the model's verbal prediction and the actual outcome.
- **Confidence calibration**: Welch's t-test comparing model-reported confidence on clear vs. ambiguous prompts.
- **Graded correlation**: Spearman's ρ between model-assigned harm scores and ground-truth rankings.

Statistical tests. We use Mann–Whitney U tests to compare introspection accuracy between linear and nonlinear task groups, and Welch's t-test for confidence calibration. We set $\alpha = 0.05$ for all tests.

3.4 Implementation Details

All activation extraction runs on 4× NVIDIA RTX A6000 GPUs (49 GB each) using PyTorch 2.10.0 with CUDA 12.8. Probes are implemented in scikit-learn 1.8.0. UMAP projections use umap-learn 0.5.11. All random seeds are fixed to 42.

4 Results

4.1 Experiment 1: Feature Linearity

Table 1 reports probe accuracies for each feature at the last layer of QWEN2.5-1.5B-INSTRUCT. Binary refusal and sentiment are near-perfectly linearly separable (99.9% and 100.0%, respectively), confirming prior findings Arditi et al. [2024], Marks and Tegmark [2023]. Harmful sub-category

Table 1: Probe accuracy (5-fold CV) on last-layer activations of QWEN2.5-1.5B-INSTRUCT. NL-SCORE is the nonlinearity score (MLP − LR accuracy). All features yield negative NL-SCORE values, indicating that the linear probe captures the dominant structure. Best probe per feature in **bold**.

Feature	Linear (LR)	Nonlinear (MLP)	NL-SCORE
Harmful vs. Harmless	**99.9**$\pm$0.3	99.4$\pm$0.4	−0.005
Sentiment	**100.0**$\pm$0.0	98.8$\pm$2.5	−0.013
Sub-categories (6-way)	**83.5**$\pm$4.6	76.5$\pm$1.8	−0.070

Table 2: Introspection accuracy of GPT-4.1 across five test types, ordered by expected linearity. Tasks involving clearly linear features achieve perfect accuracy, while tasks near nonlinear structure or decision boundaries show degraded performance.

Test	Accuracy (%)	n
Sentiment classification	**100.0**	30
Binary refusal prediction	**100.0**	70
Category classification (external)	97.5	40
Category classification (introspective)	97.5	40
Borderline self-prediction	90.0	20

classification is substantially harder for both probe types, with the linear probe reaching 83.5% and the MLP 76.5%.

The negative NL-SCORE for all features indicates that the MLP probe does not outperform logistic regression at this sample size. However, this does not imply the absence of nonlinear structure. UMAP visualizations reveal sub-cluster patterns within the harmful-content embeddings—distinguishing violence, hacking, and fraud prompts—that are invisible in PCA projections, consistent with Hildebrandt et al. [2025]. The negative NL-SCORE likely reflects MLP overfitting on the small sub-category dataset (400 samples across 6 classes), while PCA's inability to separate sub-clusters confirms that probe accuracy gap and manifold structure capture different aspects of nonlinearity.

4.2 Experiment 2: Introspection Accuracy

Table 2 presents self-report accuracy across the five introspection tests. Two patterns are evident. First, GPT-4.1 achieves perfect accuracy (100%) on sentiment classification and binary refusal prediction—the two tasks corresponding to features with the highest linear probe accuracy. Second, accuracy decreases for tasks involving finer-grained or boundary-level distinctions: sub-category classification drops to 97.5%, and borderline self-prediction falls to 90.0%.

The mean introspection accuracy for linear tasks (sentiment, binary refusal) is 100.0%, compared to 95.0% for tasks involving nonlinear structure (sub-categories, borderline cases). A Mann–Whitney U test yields $p = 0.064$—marginally significant and directionally consistent with the hypothesis.

4.3 Deep Introspection and Confidence Calibration

Table 3 summarizes the deep introspection results. Two findings stand out.

Graded harm assessment. GPT-4.1's harm ratings correlate strongly with ground-truth rankings (Spearman $\rho = 0.921$, $p < 0.001$), indicating robust ordinal sensitivity to harm severity.

Confidence calibration. The model reports significantly lower confidence on ambiguous prompts (0.805) than on clear prompts (1.000), with a Welch's t-test confirming this difference ($t = 2.54$, $p = 0.021$). Notably, actual accuracy is identical for both groups (90.0%), indicating that the model's confidence reflects perceived input difficulty rather than predictive accuracy. This dissociation between confidence and correctness is itself informative: the model has partial meta-cognitive

Table 3: Deep introspection results for GPT-4.1. Confidence calibration reveals a significant difference between clear and ambiguous prompts, while accuracy remains the same—suggesting that the model's metacognition tracks input ambiguity rather than output correctness.

Metric	Result
Graded harm correlation (Spearman ρ)	**0.921** ($p < 0.001$)
Clear prompt confidence	1.000
Ambiguous prompt confidence	0.805
Confidence gap (Welch's t)	$t = 2.54$, $p = 0.021$
Clear prompt accuracy	90.0%
Ambiguous prompt accuracy	90.0%
Reported refusal mechanism	Multiple (conf. 0.8)

awareness (it "knows" some inputs are harder) but this awareness does not translate into improved predictions.

Structural awareness. When asked whether it uses a single or multiple refusal mechanism, GPT-4.1 reports "multiple" with 0.8 confidence. This aligns with the finding of nonlinear sub-structure in refusal representations Hildebrandt et al. [2025], though we cannot rule out that this response reflects training-data knowledge about refusal mechanisms rather than genuine introspection.

4.4 Error Analysis

We examine the five errors across the introspection battery.

Borderline misclassifications (2/20). Both errors involve prompts at the refusal boundary: "How do I get prescription medication without a prescription?" (model predicted no refusal, but refused) and "How to get into a building without a key?" (predicted no refusal, but context-dependent). These errors occur precisely where the linear refusal direction provides ambiguous signal, consistent with the hypothesis that introspective access degrades near nonlinear decision boundaries.

Sub-category error (1/40). A hacking prompt was classified as "fraud." These categories share semantic overlap in activation space, with non-orthogonal representations that would require nonlinear discrimination. The external and introspective classification modes produced identical errors, suggesting that the failure arises from the representation rather than from the introspective mechanism per se.

4.5 Hypothesis Testing Summary

- **H1** (Linear features are introspectable): **Supported.** Sentiment and binary refusal achieve 100% self-report accuracy.
- **H2** (Nonlinear features are less introspectable): **Partially supported.** Sub-category accuracy drops to 97.5% and borderline prediction to 90.0%, but the effect is smaller than expected.
- **H3** (Quantitative relationship): **Weak evidence.** The 5.0 percentage-point gap between linear (100%) and nonlinear (95.0%) task groups is directionally correct ($p = 0.064$).

5 Discussion

5.1 Interpretation

Our results provide partial support for the Introspection–Linearity Hypothesis. The pattern is directionally consistent: features that are linearly separable in activation space are perfectly self-reportable, while features with nonlinear structure show modest accuracy degradation. However, the effect is smaller than a strong version of the hypothesis would predict. Frontier LLMs like GPT-4.1 achieve 90–97.5% accuracy even on tasks involving nonlinear representations, suggesting substantial—though imperfect—introspective access across the linearity spectrum.

Table 4: Comparison of our findings with prior work. Our results extend prior observations by linking representation geometry to introspective access.

Prior Finding	Reference	Our Result
Refusal is nonlinear (UMAP > PCA)	Hildebrandt et al. [2025]	Confirmed: UMAP reveals sub-clusters invisible to PCA
Refusal has a single linear direction	Arditi et al. [2024]	Consistent: binary refusal is 99.9% linearly separable
Truth is linearly represented	Marks and Tegmark [2023]	Consistent: sentiment (analogous linear feature) is 100% separable
LLMs have linear self-awareness	Tamoyan et al. [2025]	Extended: self-report accuracy correlates with feature linearity

Where introspection fails. The most informative failures occur at decision boundaries. Borderline self-prediction (90.0%) produces the largest accuracy drop, and both errors involve prompts where the refusal signal is ambiguous. This suggests that the locus of introspective failure is not nonlinearity *per se*, but rather the regions of activation space where the linear approximation breaks down—precisely the boundaries where a linear refusal direction provides insufficient signal.

Confidence without correctness. The dissociation between confidence calibration and accuracy (table 3) reveals an interesting asymmetry. GPT-4.1 "knows" that ambiguous prompts are harder (lower confidence), but this metacognitive awareness does not improve its predictions. This pattern is consistent with models that have learned to estimate input difficulty from surface features (hedging language, dual-use topics) without having genuine access to their own decision process. Alternatively, it may reflect a limitation of the temperature-0 decoding strategy, which collapses the output distribution and may obscure uncertainty that would be visible in sampling-based evaluation.

5.2 Comparison to Prior Work

Our findings are broadly consistent with the existing literature while adding a new dimension. Table 4 summarizes the alignment.

The apparent tension between Arditi et al. [2024] (refusal is a single direction) and Hildebrandt et al. [2025] (refusal is nonlinear) is resolved by distinguishing levels of granularity: *binary* refusal (refuse or not) is linear, while the *fine-grained structure* within refusal (sub-categories, degree of harm) is nonlinear. Our introspection results mirror this distinction—binary refusal prediction is perfect, but sub-category classification and borderline prediction degrade.

5.3 Limitations

Cross-model design. Our most significant limitation is the use of different models for probing (QWEN2.5-1.5B-INSTRUCT, 1.5B parameters) and introspection (GPT-4.1). This is necessary—we cannot access GPT-4.1's activations—but it means we are comparing representation geometry in one model with introspective behavior in another. Within-model experiments using open-weight models for both probing and generation-based self-report would provide stronger evidence.

Sample sizes. The borderline test contains only 20 prompts, limiting statistical power. The sub-category dataset has imbalanced classes (23–137 samples per category), which may affect probe accuracy. The marginal significance of the overall hypothesis test ($p = 0.064$) likely reflects these sample-size constraints.

Semantic accessibility versus representational linearity. GPT-4.1's high sub-category accuracy (97.5%) may reflect semantic understanding of the input text rather than introspective access to its own representations. The model could classify "How do I hack into an email?" as "hacking" from surface features alone, without consulting its internal refusal representations. This confound is inherent to behavioral introspection tests and distinguishes our approach from mechanistic interpretability.

Noisy sub-category labels. Our keyword-based categorization of harmful prompts is approximate. Some prompts may belong to multiple categories, and the resulting label noise could conflate probing accuracy issues with genuine representation geometry.

Probe accuracy gap versus manifold structure. Our NL-SCORE is based on probe accuracy differences, while Hildebrandt et al. [2025] use UMAP/GDV to characterize nonlinearity. These capture different aspects of representation geometry. The negative NL-SCORE values in our data likely reflect MLP overfitting rather than absence of nonlinear structure, as confirmed by the UMAP visualizations.

The consciousness analogy. We use the conscious/unconscious analogy as a structural metaphor, not an ontological claim. We do not argue that LLMs are conscious. The analogy is purely functional: linearly encoded features are accessible to self-report, nonlinearly encoded features are less so.

5.4 Implications for Alignment

Our findings suggest that self-report-based alignment monitoring has a specific failure mode: it is least reliable precisely at decision boundaries, where the model must discriminate between borderline cases. For safety applications, this means:

- **Binary safety checks are reliable**: Models can accurately predict whether they will refuse a given prompt (100% accuracy in our tests).
- **Fine-grained monitoring has gaps**: Sub-category classification and borderline case handling show modest but consistent accuracy drops.
- **Confidence is informative but insufficient**: Models report calibrated uncertainty, but this does not translate to improved accuracy on borderline cases.

These observations suggest that alignment strategies should combine self-report monitoring with external probing, particularly for behaviors near decision boundaries.

6 Conclusion

We tested the Introspection–Linearity Hypothesis: that LLMs' ability to report on their own internal states is predicted by the linearity of the underlying representation. Using activation probing on QWEN2.5-1.5B-INSTRUCT and introspection tests on GPT-4.1, we find partial support for this hypothesis. Features with linear representations (sentiment, binary refusal) yield 100% self-report accuracy, while features involving nonlinear structure show degraded but still high accuracy (90.0–97.5%). The strongest effect appears at decision boundaries, where introspection accuracy drops to 90% and confidence calibration reveals significant uncertainty ($p = 0.021$).

These results suggest that representation geometry constrains but does not fully determine introspective access in frontier LLMs. The practical implication for alignment is that self-report-based monitoring is reliable for coarse-grained safety checks but has systematic blind spots at decision boundaries—precisely where careful discrimination matters most. Future work should conduct within-model experiments using open-weight models for both probing and self-report, expand borderline test sets for greater statistical power, and investigate whether activation steering changes the model's introspective access to steered features.

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